

AE4889
Special Topics in Astrodynamics
MIMOSA – MIssion to Mars with a sOlar SAil

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github.com/dptrias/SpecialTopicsAstrodynamics

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Personal problem specification

Parameter	Value
Sun–Earth Lagrange point	L_2
Sun–Mars Lagrange point	L_2
Shift in Lagrange points	0.0005
Radius Martian parking orbit	306 666 km
Lightness number	0.025

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List of Acronyms

AEP Artificial Equilibrium Point. [2](#)

CR3BP Circular Restricted Three-Body Problem. [1](#)

JPL Jet Propulsion Laboratory. [1](#)

WP1 Sun–Earth and Sun–Mars Lagrange points

1.1a The Lagrange points of the Sun–Earth and Sun–Mars systems are computed using Newton’s method over equation (1), which is derived from the [Circular Restricted Three-Body Problem \(CR3BP\)](#). The resulting dimensionless coordinates of the colinear Lagrange points (L_2) are presented in Table 1.

$$x - \frac{1 - \mu}{(x + \mu)^2} - \frac{\mu}{(x - 1 + \mu)^2} = 0 \quad (1)$$

Table 1: Dimensionless coordinates (x_i, y_i, z_i) of the Sun–Earth and Sun–Mars colinear Lagrange points in their respective synodic reference frames, obtained from the solution of the CR3BP using a Newton method.

Lagrange point	x_i [LU]	y_i [LU]	z_i [LU]
Sun–Earth L_2	1.01009044	0	0
Sun–Mars L_2	1.00476311	0	0

1.1b For verification purposes, the dimensionless coordinates of the colinear Lagrange points are also extracted from the [Jet Propulsion Laboratory \(JPL\) Solar System Dynamics three-body periodic orbits tool](#), which are presented in Table 2. It can be observed that for the represented number of digits, the values are identical, with differences on the order of 10^{-9} LU, likely due to shared system parameters. *(50 words)*

Table 2: Dimensionless x coordinates of the Sun–Earth and Sun–Mars colinear Lagrange points in their respective synodic reference frames per JPL and their absolute differences w.r.t. the values obtained from the Newton method.

Lagrange point	x_i [LU]	Δx_i [LU]
Sun–Earth L_2	1.01009044	$4.21585256 \times 10^{-9}$
Sun–Mars L_2	1.00476311	$3.32491457 \times 10^{-9}$

1.2 A greater value of μ means the primary’s gravitational pull weakens ($1 - \mu$) while the secondary’s strengthens (μ). Since L_2 sits close to the secondary, this increase is more strongly felt there and must be offset by moving the equilibrium point further from it. Increasing this distance has two compensating effects: it reduces the secondary’s pull more sharply than the primary’s (due to proximity), and it increases the centrifugal acceleration, which opposes both gravitational contributions. The net result is that a more massive secondary, as in the Sun–Earth system, pushes L_2 further from the planet than in the Sun–Mars case. *(100 words)*

WP2 Artificial Lagrange points

2.1a See Table 3.

Table 3: Required lightness number to achieve the specified shift in the Lagrange points.

Lagrange point	Shift in Lagrange point along the Sun–Planet line [LU]	Required lightness number, β [-]
Sun–Earth L_2	0.0005	$4.76757628 \times 10^{-3}$
Sun–Mars L_2	0.0005	$5.06104189 \times 10^{-3}$

With respect to feasibility, NASA’s *ACS3* (currently in orbit), based on DCB deployable composite boom technology, has a lightness number $\beta = 0.025$ [1], an order of magnitude above the required values. This confirms that the artificial Lagrange point shifts studied are achievable with current solar sail technology. (*47 words*)

2.1b Since the required shift is along the Sun–planet line, and all forces at the collinear Lagrange points are aligned along this axis, the sail orientation needed to achieve the desired shift is likewise along this line. As the spacecraft moves sunward from L_2 , the gravitational pulls of both bodies strengthen while the centrifugal acceleration diminishes, requiring a compensating force directed away from the Sun. This uniquely determines the sail orientation, with the sail normal away from the Sun. (*80 words*)

Table 4: Required sail orientation to achieve the required shift in the Lagrange points.

Lagrange point	$n_{x,i}$	$n_{y,i}$	$n_{z,i}$
Sun–Earth L_2	1	0	0
Sun–Mars L_2	1	0	0

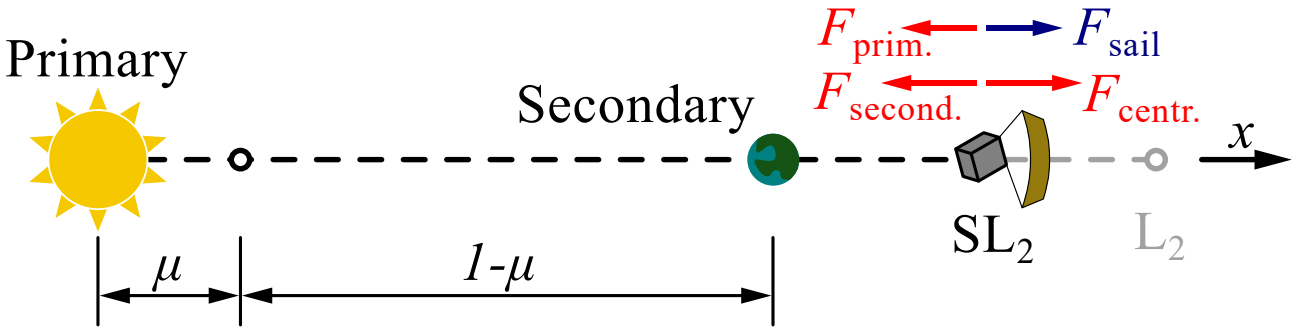


Figure 1: Schematic of all acceleration vectors acting on the spacecraft in the solar sail augmented CR3BP.

2.2 To verify the required lightness numbers calculations the colinear [Artificial Equilibrium Point \(AEP\)](#) is calculated using this work’s implementation and compared to the value reported in [2] for the Sun–Earth system with a AEP located at $x_{\text{AEP}} = 0.983867$ LU. The resulting lightness number is $\beta = 0.0363$, which matches the value reported in [2], confirming the correctness of the implementation. (*61 words*)

2.3 The required area of the solar sail is derived from the definition of the lightness number

and the expression for the solar radiation pressure:

$$A = \frac{\beta \mu_{\odot} m}{2r^2} \frac{c}{S_{\odot}} \left(\frac{r}{\text{AU}} \right)^2 \quad (2)$$

where the value of the solar constant, $S_{\odot}$, is taken from [3].

The resulting sizes required to achieve the specified shifts in the Lagrange points are shown in Table 5.

Table 5: Required solar sail side-length to achieve the required shift in the Lagrange points.

Lagrange point	Required solar sail side-length [m]
Sun–Earth L_2	17.6459
Sun–Mars L_2	18.1809

These sizes are at the upper limit of currently feasible solar sails. The maximum sail size of DCB-based designs in CubeSat form factor is 16.5 m, with larger platforms expected to be capable of deploying sails of up to 22.8 m (*27U HIPERSail*) [1].

2.4 To analyse the sensitivity of the required lightness number to the shift in the Lagrange point, a range of 21 shifts from 0.0005 LU to 0.001 LU is considered, and the corresponding lightness numbers are computed using the expression from Task 2.1a. The results are shown in Figure 2.

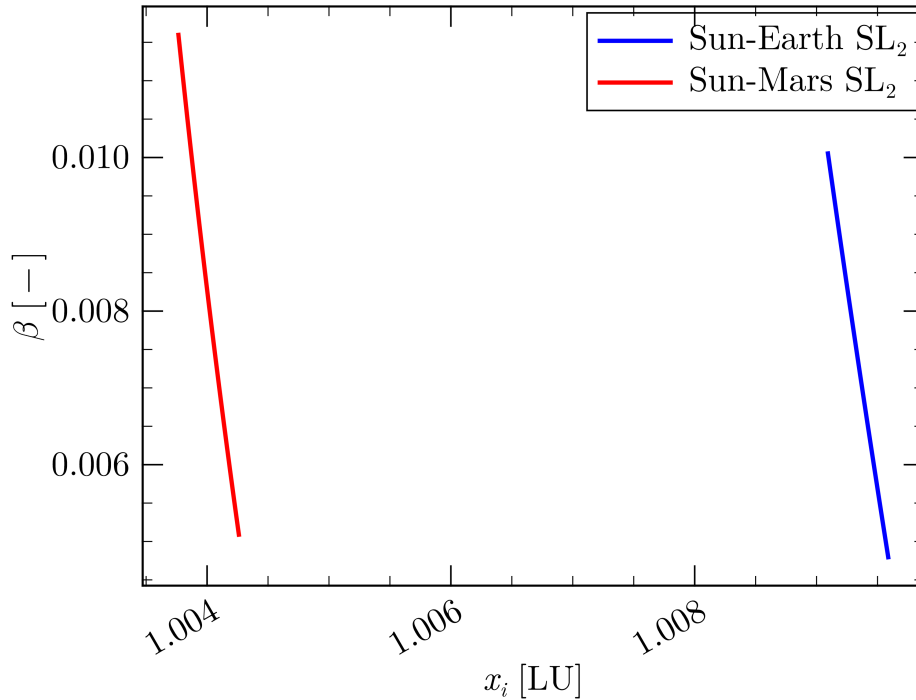


Figure 2: Sensitivity of the required lightness number, β , to the shift in the Lagrange point, Δx , for the Sun–Earth and Sun–Mars systems.

As the equilibrium position moves closer to the planet, the gravitational attraction from both bodies strengthens while the centrifugal acceleration weakens. To counterbalance these effects, the sail acceleration must increase, requiring a larger lightness number. The required β is observed to follow a near-linear increase with decreasing distance from the Sun for both systems. (*54 words*)

WP3 Stability of the (artificial) Lagrange points

3.1a For the linearization constant sail orientation in the rotating frame is assumed, i.e. $\hat{\mathbf{n}} = \text{constant}$. The eigenvalues at the equilibrium points are given in Table 6.

Table 6: Eigenvalues of the classical and artificial Lagrange points.

Lagrange point	λ_1	λ_2	λ_3	λ_4
Sun–Earth Classical	−2.484281	−2.056992 <i>i</i>	+2.056992 <i>i</i>	+2.484281
Sun–Earth Artificial	−2.673872	−2.174388 <i>i</i>	+2.174388 <i>i</i>	+2.673872
Sun–Mars Classical	−2.496888	−2.064657 <i>i</i>	+2.064657 <i>i</i>	+2.496888
Sun–Mars Artificial	−2.931888	−2.335453 <i>i</i>	+2.335453 <i>i</i>	+2.931888

3.1b From the linearization of the problem around the equilibrium points it can be observed that all cases have a positive real eigenvalue, indicating that the nonlinear system is unstable around the equilibrium point [4]. All eigenvalues take similar values across the 4 cases, with slightly higher magnitudes in the Sun–Mars system, indicating faster local dynamics. The introduction of the solar sail also increases the magnitude of all eigenvalues slightly, but not enough to have a qualitative effect on the stability of the system. (82 words)

3.2 For verification purposes the eigenvalue calculation tool results are compared against values presented in [5] for a value of $\mu = 3 \times 10^{-6}$. The resulting eigenvalues are shown in Table 7.

Table 7: Verification of the eigenvalue calculation tool for $\mu = 3 \times 10^{-6}$.

Source	λ_1	λ_2	λ_3	λ_4
This work	−2.4844225586	−2.0570784929 <i>i</i>	+2.0570784929 <i>i</i>	+2.4844225586
Szebehely [5]	−2.4844225586	−2.0570784929 <i>i</i>	+2.0570784929 <i>i</i>	−2.4844225586

WP4 Invariant manifolds and application to Mars exploration mission

4.1 See Figure 3

4.2a The unstable manifold departing L_2 intersects the [Mars Parking Orbit \(MPO\)](#), enabling a propulsion-free transfer. However, the manifolds do not intersect Earth’s orbit, as it lies on a different energy level, meaning propulsion is required for the Earth to Sun–Mars L_2 leg. (40 words)

4.2b See Table 8 and Figure 4.

Table 8: ΔV s required to establish a natural transfer between Earth and Mars.

ΔV_E [m/s]	ΔV_M [m/s]	ΔV_{TOT} [m/s]
–	294.0023	–

4.3 See Table 9 and Figures 5 and 6.

Table 9: Details of the solar sail Earth–Mars transfer with the smallest total ΔV , ΔV_{TOT} .

Leg 1 optimal cone angle [°]	Leg 2 optimal cone angle [°]	ΔV_E [m/s]	ΔV_M [m/s]	ΔV_{TOT} [m/s]	Time of Flight Leg 1 [Earth years]	Time of Flight Leg 2 [Earth years]
42.5	-67.5	611.6762	276.6996	888.3757	4.0092	0.5521

4.4 Comparing the results presented here with those obtained by Cala González Penagos (student #5484944), it can be observed that González’s solution achieves a lower ΔV by approximately 100 m/s, at the expense of an increase in time of flight of about one year. This difference arises mainly in Leg 1. González’s trajectory spans approximately 270° around the Sun (in the rotating frame), whereas the trajectory obtained in this work covers about 180° . The longer heliocentric path suggests a more gradual energy exchange with the solar sailing dynamics, reducing ΔV but increasing flight time.

This difference is likely linked to the chosen arrival point as González targets the L_1 point, while this work targets L_2 . The different dynamical environments at these points lead to distinct optimal transfer geometries, with the arrival condition appearing to be the main driver of the observed trade-off, as the lightness numbers differ only marginally (0.005).

In Leg 2, an L_1 departure yields both lower ΔV and a time of flight approximately four times shorter than the L_2 case. However, this improvement is insufficient to offset the substantially longer duration of Leg 1, resulting in a higher overall time of flight for González’s trajectory. (200 words)

4.5 The earliest possible departure date for the optimal transfer is determined by the synodic mean anomaly of the Earth and Mars as follows:

$$t_{\text{dep}} = \frac{\text{mod}(\theta_{\text{dep}} - \theta_0, 2\pi)}{n_E - n_M} + t_0 \quad (3)$$

The earliest departure date is found to be on the **25th of May, 2030**.

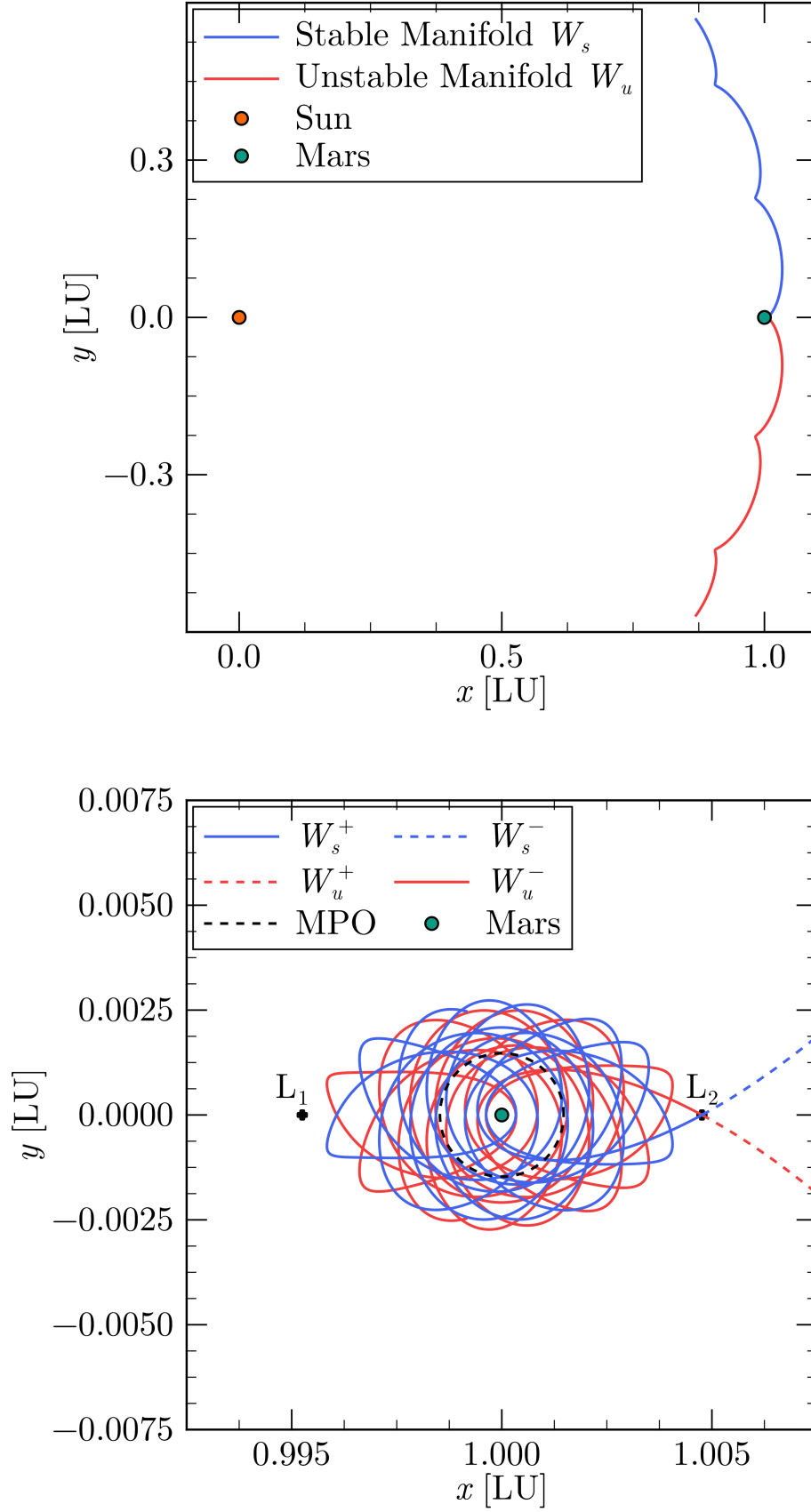
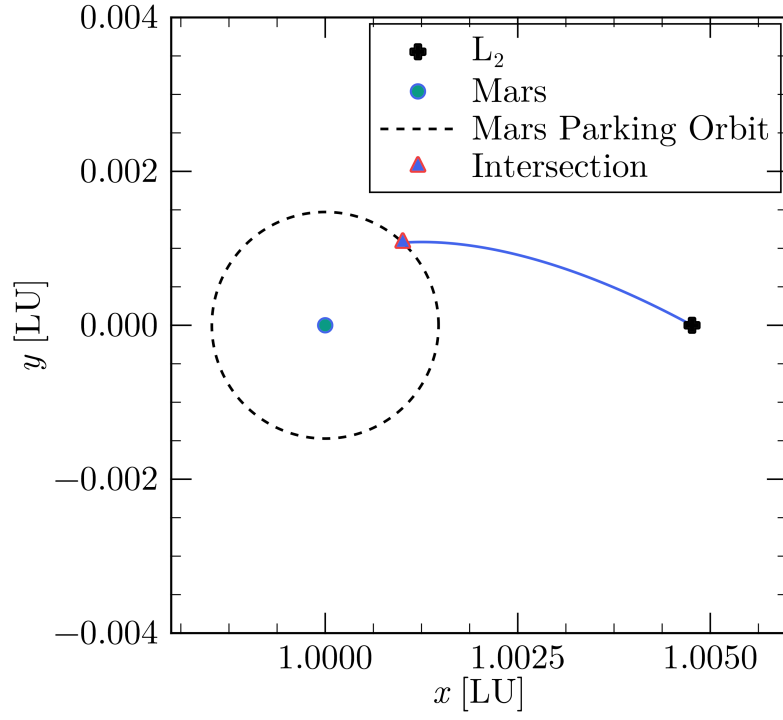
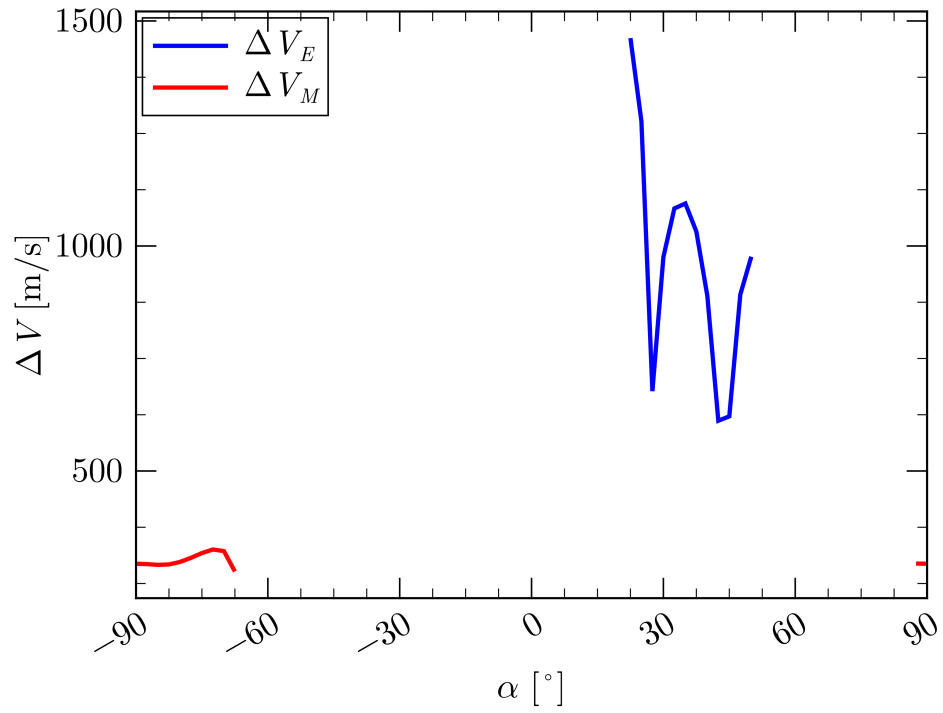


Figure 3: Stable(W^s) and unstable(W^u) invariant manifolds of the classical L_2 point in the Sun-Mars system.

Figure 4: Close up of the natural transfer from L_2 to the MPO.Figure 5: ΔV s of the solar sail transfer as a function of the cone angle.

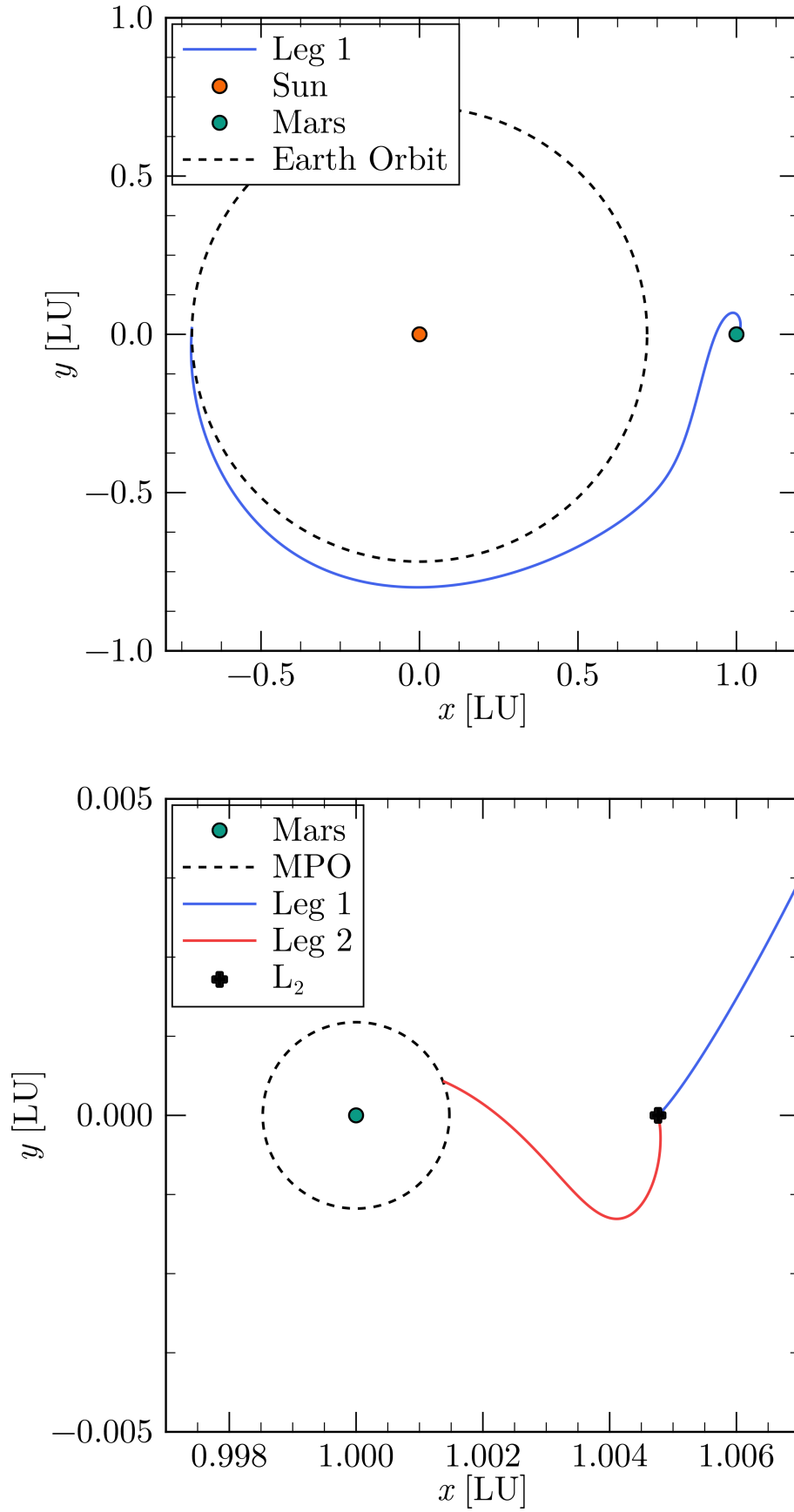


Figure 6: Optimal transfer trajectory from Earth to the MPO via the Sun–Mars L_2 point using a solar sail.

4.6 The results demonstrate that solar sail missions from Earth to Mars with minimized ΔV are feasible. However, several limitations constrain practical applicability. The assumed 100 kg spacecraft is already at the boundary of current solar sail technology, and the true useful payload is further reduced by the mass of sail support systems, attitude control hardware, propulsion systems for the required ΔV maneuvers, and Mars descent mechanisms, none of which have been accounted for in this study. This makes the small payload the primary limiting factor for the exploitation of this technology in the near term.

A viable niche may nonetheless emerge in the long term. In the short term, Mars settlement will rely primarily on in-situ resource utilization [6], but once sustained settlements with predictable long-term supply demands are established, low-cost propellant-efficient delivery architectures of this kind could play a meaningful supporting role. *(144 words)*

References

- [1] W. Wilkie, J. Fernandez, O. Stohlman, *et al.*, “An overview of the NASA Advanced Composite Solar Sail (ACS3) technology demonstration project,” Jan. 2021.
- [2] J. Heiligers, B. Diedrich, B. Derbes, and C. McInnes, “Sunjammer : Preliminary end-to-end mission design,” in *AIAA/AAS Astrodynamics Specialist Conference 2014*, USA, Aug. 2014. [Online]. Available: <https://strathprints.strath.ac.uk/49051/>.
- [3] G. Kopp and J. L. Lean, “A new, lower value of total solar irradiance: Evidence and climate significance,” *Geophysical Research Letters*, vol. 38, no. 1, DOI: <https://doi.org/10.1029/2010GL045777>.
- [4] F. Verhulst, *Nonlinear Differential Equations and Dynamical Systems* (Universitext), 2nd ed. Berlin, Heidelberg: Springer-Verlag, 1996, ISBN: 978-3-540-60934-6.
- [5] V. G. Szebehely, *Theory of Orbits: The Restricted Problem of Three Bodies*. New York: Academic Press, 1967, ISBN: 9780123957320.
- [6] M. N. Zamani, M. S. Ahmad Shah, S. Umar, *et al.*, “The feasibility of in-situ resource utilisation binder systems for construction materials on mars: A review,” *Advances in Space Research*, vol. 74, no. 3, pp. 1535–1561, 2024, ISSN: 0273-1177. DOI: <https://doi.org/10.1016/j.asr.2024.04.059>.

A AI Statement

AI was used throughout this work for assistance with *Julia* plotting.

How to do a 1x2 figure in julia

27 May

In Makie (assuming you're using that), a 1x2 figure is straightforward:

```
julia
fig = Figure(size = (1200, 500))

ax1 = Axis(fig[1, 1], xlabel = "x", ylabel = "y", title = "Left")
ax2 = Axis(fig[1, 2], xlabel = "x", ylabel = "y", title = "Right")

# plot into each axis
lines!(ax1, x, y1)
lines!(ax2, x, y2)
```

fig

The fig[row, col] indexing is all you need. A few useful extras:

Equal column widths are the default, but you can control spacing:

```
julia
colgap!(fig.layout, 20) # gap in pixels between columns
Shared axis limits:
```

```
julia
linkxaxes!(ax1, ax2)
linkyaxes!(ax1, ax2)
Overall figure padding:
```

```
julia
fig = Figure(size = (1200, 500), figure_padding = 10)
Saving:
```

```
julia
save("output.pdf", fig)
That's the core of it. The fig[1, 1] and fig[1, 2] grid positions
    handle the layout automatically.
```

im now using Inf for a variable im plotting, how do i filter them for plotting

27 May

A simple boolean mask:

```
julia
valid = isfinite.(y_values_with_inf)
Then plot with:
```

```
julia
lines!(ax, x_values[valid], y_values_with_inf[valid])
```

how to make a legend box not transparent

27 May

In Makie:

```
julia
axislegend(ax, backgroundcolor = :white)
Or if you are using a figure-level legend:
```

```
julia
Legend(fig[1, 2], ax, backgroundcolor = :white)
You can also control the border:
```

```
julia
axislegend(ax, backgroundcolor = :white, framecolor = :black,
           framewidth = 1)
have it cover the axis ticks
```

27 May

Set a high tellwidth/tellheight and use framevisible plus depth ordering... actually in Makie the simplest way is just to place the legend manually with a high z-order isn't directly exposed, but you can control it with:

```
julia
axislegend(ax, backgroundcolor = :white, framecolor = :black,
           framewidth = 1, margin = (10, 10, 10, 10))
```

If it is still going behind the ticks, switch to a Legend placed inside the axis using halign and valign:

```
julia
Legend(fig[1, 1], ax, backgroundcolor = :white, framecolor = :black,
       framewidth = 1, tellwidth = false, tellheight = false,
       halign = :right, valign = :top, margin = (10, 10, 10, 10))
```

The tellwidth = false, tellheight = false is the key part, it lets the legend float inside the axis panel rather than being pushed to a grid cell, and it will naturally render on top of the axis content including ticks.

simpler

27 May

```
julia
axislegend(ax, position = :rt, backgroundcolor = :white)
:rt is right-top. Other options are :lt, :lb, :rb for the four corners.
```